

Additional marking guidance for 8(a)(ii)

Method 1

$$hf = 4.37 \times 10^{-19} \text{ J} \quad (1)$$

$$\text{For Barium} \quad hf - \phi = 0.34 \times 10^{-19} \text{ J} \quad (1)$$

$$V_{\text{stop}} = 0.21 \text{ V} \quad (1)$$

Therefore barium (1)

Method 2

$$hf = 4.37 \times 10^{-19} \text{ J} \quad (1)$$

$$h\phi < 4.37 \times 10^{-19} \text{ J or not Ca or Zn} \quad (1)$$

$$\text{For potassium, Either } hf - \phi = 0.69 \times 10^{-19} \text{ J} \quad (1)$$

$$\text{or } V_{\text{stop}} = 0.43 \text{ V} \quad (1)$$

Therefore barium (1)

Method 3

$$hf = 4.37 \times 10^{-19} \text{ J} \quad (1)$$

$$h\phi < 4.37 \times 10^{-19} \text{ J or not Ca or Zn} \quad (1)$$

$$E_k < 0.56 \times 10^{-19} \text{ J or } \phi > 3.81 \times 10^{-19} \text{ J} \quad (1)$$

[Don't penalise $E_k = 0.56 \times 10^{-19} \text{ J}$ or $\phi = 3.81 \times 10^{-19}$ if seen]

Therefore barium (1)